Course Authoring

- Instructor Onboarding: Course creation style is asked for future AI recommendation, ai feature customization to understand goals of teacher using template options
- Upload docs, PDFs, slides, video links, article links, website links, quizlet, drive authorization for google docs, notes app → AI drafts full course structure
- All source types feed the same RAG namespace - AI drafts from everything together
- Video links auto-transcribed; transcription feeds RAG so AI can reference what was said
- Instructor crops videos to specific timestamp ranges and embeds only the relevant clip
- Platform suggests related videos during the editing process
- Visual drag-and-drop editor with real-time preview (Final Cut model)
- AI draft is always editable - instructor approves every block before publish
- Concept staging with AI-suggested prerequisite graph
- Three explanation modes per concept - teacher chooses which to implement

Content Blocks

- AI-generated Socratic dialogue
- Raw source excerpts with highlighting
- Manim + Claude animation pipeline (type a description → get a 3Blue1Brown-style animation in 30 seconds)
- Interactive simulation primitives (sliders, particle systems, 3D viewers, state diagrams)
- Cropped video embed with timestamp control
- Article/website embed
- MCQ, scenario, ranking, AI-critique, open response
- Reflection prompts
- Flashcard creation
- Quizzes

Student Experience

- Learning style onboarding (age, goals, preferred explanation mode, session length)
- Switchable explanation modes per concept - tap an icon to change view (🌍 real world examples, 💡 conversational/informal, 📖 textbook style summary) depending on whichever version is available that the instructor allowed to generate (unimplemented modes grayed out, student's onboarding preference is the default)
- A different icon that the student can click that shows if an interactive mode (animation, simulation, activity) for this module is available; this icon is greyed out if it doesn't exist
- Persistent AI assistant sidebar available on every screen throughout the course

- Assistant is context-aware - knows which concept, block, and explanation mode the student is viewing
- Assistant can surface past reflections and connect current concept to previous learning
- Assistant is guardrailed - won't give direct quiz answers, redirects to hints
- 4-level progressive hint system (nudge → concept → direction → explanation)
- Socratic tutor that never gives the answer first (separate from the assistant - tutor drives the lesson, assistant is on-demand)
- Mandatory reflection after every session
- Personalized next-concept recommendation based on actual mastery
- As the student is watching the video , there should be an Ai chat assistant present in the side where the student can ask questions, and the AI answers the question in the chat and offers an option to go back to a specific timestamp where the answer lies and actually answers it
- Hand designed, cartoonish features on UI to make it feel like a friendly experience so it feels more natural , like animated features on Tolan

Adaptive Engine

- Bayesian Knowledge Tracing per concept (5 states: Not Seen → Exposed → Practiced → Understood → Mastered)
- Misconception detection and logging
- Planner Agent sequences concepts per learner based on prerequisite graph + mastery state
- Session adapts in real time - if a student is struggling, the agent shifts approach by reinforcing missed content/it will let the student know what they have been messing up on, and accordingly assign questions in any way (through animation, flashcards, etc) to improve that specific concept
- Explanation mode switch patterns tracked and fed back to teacher dashboard

Agentic Architecture

- Orchestrator Agent - routes interactions, detects intent, selects explanation mode
- Concept Tutor Agent - Socratic teaching, retrieves from course RAG, adapts to selected explanation mode
- Quiz Coach Agent - generates questions dynamically, updates BKT mastery, logs misconceptions
- Reflection Agent - prompts metacognition, probes shallow responses, stores reflection embeddings
- Planner Agent - reads Learner Model, sequences via prerequisite graph, generates next-session card, generates structural suggestions for teacher dashboard
- Student Assistant Agent - persistent sidebar, context-aware, guardrailed, on-demand help without spoiling answers

Teacher Dashboard

- Concept mastery heatmap (green/yellow/red across all learners)
- Misconception log ranked by frequency with one-click edit to fix the block
- Question difficulty analysis (flags too easy or too hard)
- Explanation mode usage tracking (which modes students switch away from)
- Engagement timeline (where learners drop off, skip, or stall)
- AI-generated suggestions for structural changes
- Video suggestion panel - platform recommends related videos the instructor can embed
- Real-time updates - fix a problem before the next cohort hits it

Course Self-Improvement

- Closed feedback loop: learner data → teacher dashboard → editor → republish
- AI suggests making concepts optional if skip data shows no downstream impact
- Mode switch patterns reveal which explanations aren't working
- Video transcription enriches RAG over time - assistant and tutor get smarter as more source material is added
- Course gets better every week without rebuilding